

ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH  
TRƯỜNG ĐẠI HỌC BÁCH KHOA  
KHOA KHOA HỌC VÀ KỸ THUẬT MÁY TÍNH

Vietnam National University – Ho Chi Minh City · Ho Chi Minh City University of Technology  
Faculty of Computer Science and Engineering



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Programming Foundations for Data Analysis and Visualization

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ASSIGNMENT REPORT · PART 2 OF 3 · TEXT DATA

# Vietnamese News Topic Classification

*A self-crawled VnExpress dataset, bag-of-words models and multilingual sentence embeddings*

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Lecturer **Dr. Lê Thành Sách**  
Group **G3**  
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| No. | Full name            | Student ID | Main contribution                                               |
|-----|----------------------|------------|-----------------------------------------------------------------|
| 1   | Ly Minh Trung        | 2570349    | Tabular and time-series analysis; website design and deployment |
| 2   | Dinh Truong Tue Linh | 2570441    | Text analysis (Vietnamese news classification)                  |
| 3   | Nguyen Hoang Nam     | 2570261    | Text analysis (Vietnamese news classification)                  |

[Project page](#) · [Notebook on GitHub](#) · Video: to be published

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# 1 Problem and objective

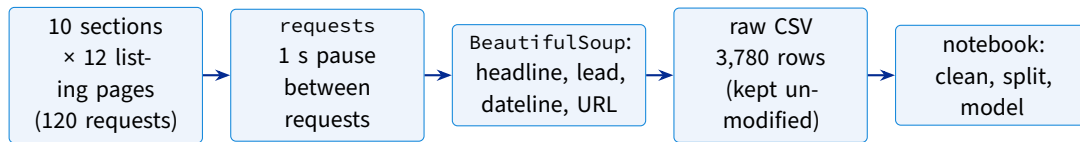
News websites tag every article with a section so that readers can browse, search and receive recommendations. Given only the **headline** and the **lead paragraph** (“sapo”) of a Vietnamese news article, we predict which of 10 newspaper sections it belongs to: News (Thời sự), World, Business, Sci-Tech, Entertainment, Sports, Law, Education, Health and Travel.

**Objective.** Build a topic classifier for short Vietnamese texts, compare classical bag-of-words models with a pretrained multilingual sentence-embedding model, and understand which mistakes remain.

**How success is measured.** **Macro-F1** — the mean of the per-section F1 scores, so every section counts equally — plus accuracy, on a held-out test set used once.

## 2 Dataset: collected by the group

No ready-made dataset was used: the group wrote a crawler — `crawl_vnexpress.py` in `text/crawler/`, built on `requests` and `BeautifulSoup` — for *VnExpress* [1], one of Vietnam’s largest online newspapers, and ran it on 27 September 2026.



**Figure 1.** Data collection pipeline. Only headlines, leads and URLs are stored (no article bodies), for non-commercial coursework.

**Table 1.** Fields of the crawled dataset.

| Column                                                | Meaning                                                                        |
|-------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>title</code>                                    | headline                                                                       |
| <code>description</code>                              | lead paragraph (sapo)                                                          |
| <code>location</code>                                 | dateline such as “Hà Nội” or “Indonesia”, when present (not used as a feature) |
| <code>url</code>                                      | article URL — unique key                                                       |
| <code>category_slug</code> ,<br><code>category</code> | section = <b>label</b>                                                         |
| <code>page</code> , <code>crawled_at</code>           | listing page number and crawl time                                             |

**Table 2.** Assignment requirements for text data (section 4.3) and how this dataset meets them.

| Requirement                | This dataset                                           | Status     |
|----------------------------|--------------------------------------------------------|------------|
| At least 2,000 samples     | 3,780 articles (3,769 after cleaning)                  | <b>Met</b> |
| Vietnamese data encouraged | all headlines and leads are Vietnamese                 | <b>Met</b> |
| Self-collected data valued | crawled by the group; the crawler is in the repository | <b>Met</b> |

### 3 Exploratory data analysis

**Quality.** 24 articles have no lead paragraph (mostly photo or video stories); 7 URLs appear twice within the same section (an article was pinned while the listing shifted between requests); 2 URLs are listed under two different sections, which makes their label ambiguous. 0 headlines needed Unicode normalisation.

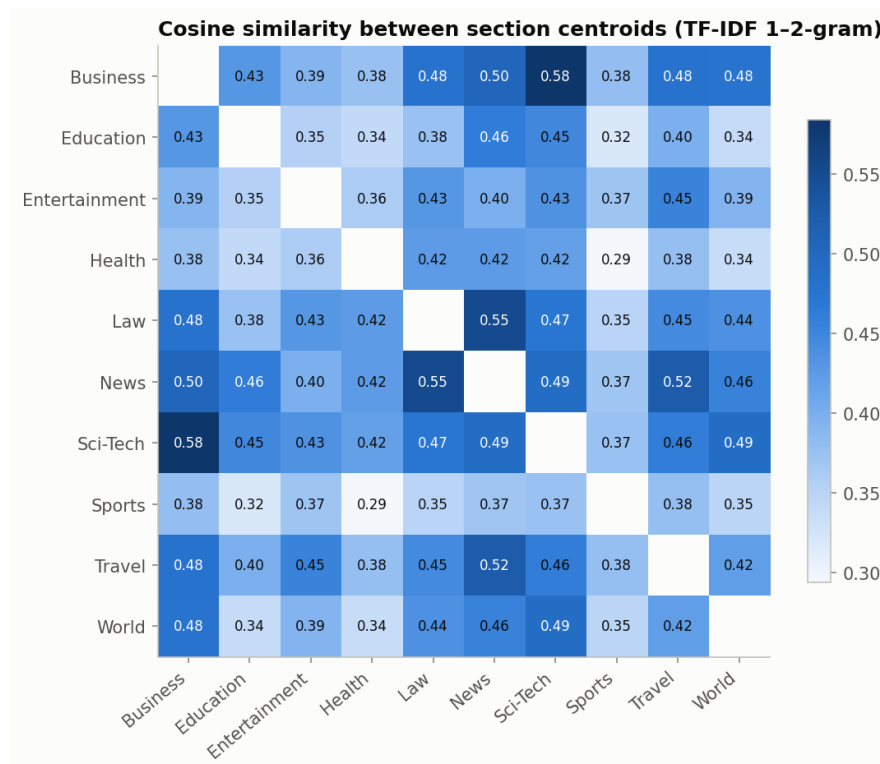
**Balance and length.** The crawler read the same number of pages per section, so the classes are balanced (374–391 articles each). Texts are short: a median headline has 12 syllables and a median lead 31 — about the length of one tweet. Vietnamese separates *syllables*, not words, with spaces (“bóng đá”, football, is two tokens), which shapes the choice of representation (Section 4).

**Table 3.** The eight n-grams most over-represented in each section ( $\chi^2$  test of each TF-IDF 1–2-gram, section vs. rest, training set).

| Section       | Top 8 discriminative n-grams ( $\chi^2$ , training set)                 |
|---------------|-------------------------------------------------------------------------|
| Business      | lãi, phiếu, doanh nghiệp, cổ phiếu, lãi suất, doanh, khoán, chứng khoán |
| Education     | học, đại học, sinh, trường, học sinh, sinh viên, đại, ngành             |
| Entertainment | phim, miss, diễn viên, miss world, diễn, hoa hậu, ca, hậu               |
| Health        | bệnh, huyết, bác sĩ, máu, viêm, ăn, uống, ung                           |
| Law           | bị cáo, cáo, bị, cáo buộc, công an, sát hại, tù, lừa                    |
| News          | ngập, mưa, mưa lớn, sông, quốc lộ, lộ num, sạt, lở                      |
| Sci-Tech      | iphone, robot, num pro, pro, iphone num, apple, ai, robot hình          |
| Sports        | cup, hlv, asiad, man, trận, asiad num, league, fifa                     |
| Travel        | khách, du, du khách, du lịch, điểm đến, khách việt, lịch, dịp           |
| World         | trump, iran, ukraine, ông trump, tổng thống, tổng, nga, houthi          |

**Question.** Which words does each section “own”?

**Finding.** Two kinds of signal appear. **Stable topic vocabulary:** *Health* — *bệnh, bác sĩ, viêm*; *Education* — *học sinh, đại học, sinh viên*; *Law* — *bị cáo, cáo buộc, tù*; *Business* — *cổ phiếu, lãi suất, chứng khoán*. Bigrams such as *cổ phiếu* and *bác sĩ* rank next to their single syllables, so syllable pairs recover real words. **Time-specific names:** *Sci-Tech* is driven by *iphone, apple, robot*, *World* by *trump, iran, ukraine*, *Sports* by *asiad, fifa*, and *News* by recent floods (*ngập, mưa lớn, sạt lở*). These reflect the news cycle at crawl time rather than the section itself — a risk discussed in the limitations.



**Figure 2.** Cosine similarity between the average TF-IDF vector (centroid) of each section.

**Question.** Before training any model, can we predict which sections will be confused?

**Finding.** The most similar pairs are **Business and Sci-Tech** (0.58) — chips, technology companies and market value appear in both — and **Law and News** (0.55) — police and accidents appear in both. *Sports* is the most isolated section. Section 7 checks this prediction against the confusion matrix.

## 4 Text preparation

**Table 4.** Preparation steps.

| Step                                | What is done and why                                                                                                                                                                                                            |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duplicates                          | keep one copy of URLs repeated within a section; drop URLs listed under two sections (ambiguous label) → 3,769 articles (11 rows removed)                                                                                       |
| Missing values                      | a missing lead becomes an empty string, so the article is classified from its headline alone                                                                                                                                    |
| Input text                          | headline + “ . ” + lead                                                                                                                                                                                                         |
| Cleaning<br>(VietnameseTextCleaner) | Unicode NFC normalisation (the same diacritic can be stored as one or two code points); lower-casing; numbers → token num; punctuation removed                                                                                  |
| Tokens                              | pattern <code>\b\w+\b</code> keeps one-character syllables; the scikit-learn default would silently drop 27 of them, including <code>ô</code> (in <code>ô tô</code> , <code>car</code> ) and <code>ở</code> ( <code>at</code> ) |
| Label encoding                      | <code>LabelEncoder</code> : 10 sections → 0–9                                                                                                                                                                                   |
| Split                               | stratified 70 / 15 / 15: 2,638 / 565 / 566 articles                                                                                                                                                                             |

For text, *encoding* means turning each document into a vector and *scaling* means weighting and normalising that vector. Four representations are compared, all fitted on the training set only:

**Table 5.** Text representations compared.

| Representation   | Idea                                                | Weighting / scaling   | Vocabulary |
|------------------|-----------------------------------------------------|-----------------------|------------|
| BoW, 1-gram      | how often each syllable occurs                      | raw counts            | 2,950      |
| TF-IDF, 1-gram   | counts re-weighted by rarity across documents       | sublinear tf, L2 norm | 2,950      |
| TF-IDF, 1–2-gram | adds syllable pairs → compound words                | sublinear tf, L2 norm | 15,517     |
| TF-IDF, char 2–5 | character n-grams inside words → robust to spelling | sublinear tf, L2 norm | 19,160     |

## 5 Modelling

### 5.1 Representation × classifier grid

Each representation is paired with four classifiers — Multinomial Naive Bayes, Logistic Regression, a linear Support Vector Machine and k-nearest neighbours with cosine distance — and scored on the validation set (TextBenchmark class).

**Table 6.** Validation macro-F1 for every representation × classifier pair (best in bold). A stratified random guess scores 0.081.

| Representation  | Multinomial NB | Logistic Reg. | Linear SVM | kNN (cosine) |
|-----------------|----------------|---------------|------------|--------------|
| BoW 1-gram      | 0.839          | 0.820         | 0.819      | 0.751        |
| TF-IDF 1-gram   | 0.837          | 0.848         | 0.858      | 0.837        |
| TF-IDF 1–2-gram | 0.856          | <b>0.864</b>  | 0.863      | 0.849        |
| TF-IDF char 2–5 | 0.825          | 0.847         | 0.858      | 0.819        |

**Question.** Which representation and classifier work best for short Vietnamese texts?

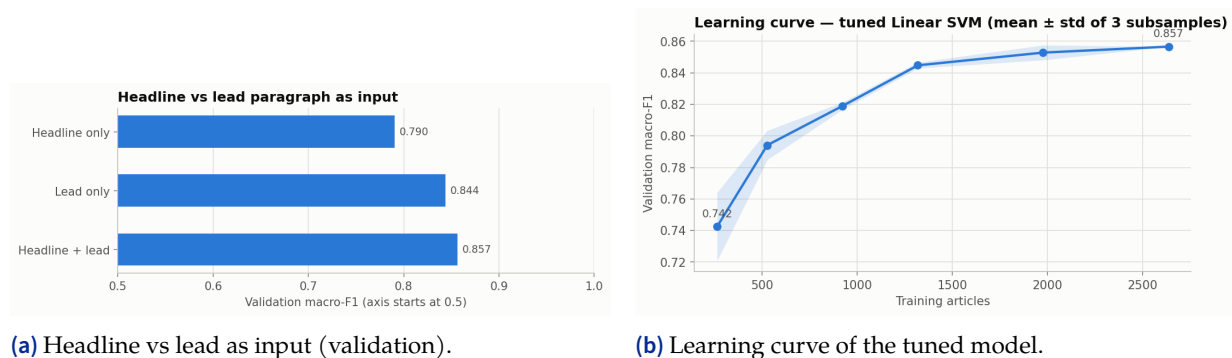
**Finding.** **TF-IDF beats raw counts** for the discriminative models — Logistic Regression +0.028, Linear SVM +0.039, kNN +0.086 — because IDF down-weights syllables that occur everywhere (*của, là, và*), acting as an automatic stop-word list; Naive Bayes, designed for counts, does not gain. **Adding bigrams helps every model** (+0.005 to +0.019), while character n-grams do not beat word bigrams. The best pair is TF-IDF 1–2-gram + Logistic Regression (0.864); the Linear SVM on the same features is practically tied (0.863) and trains several times faster.

### 5.2 Tuning

The TF-IDF + Linear SVM pipeline was tuned with 5-fold stratified cross-validation on the training set over the n-gram range, `min_df`, sublinear term frequency and the regularisation constant `c`. The best setting uses 1–3-grams, `min_df` = 1 and `c` = 3 (cross-validated macro-F1 0.865). On the single validation split it scores 0.857 — within about 0.01 of the untuned models, i.e. a handful of the 565 articles; the 5-fold estimate is the more reliable of the two.

## 6 Extra experiments

### 6.1 Which field carries the signal, and would more data help?



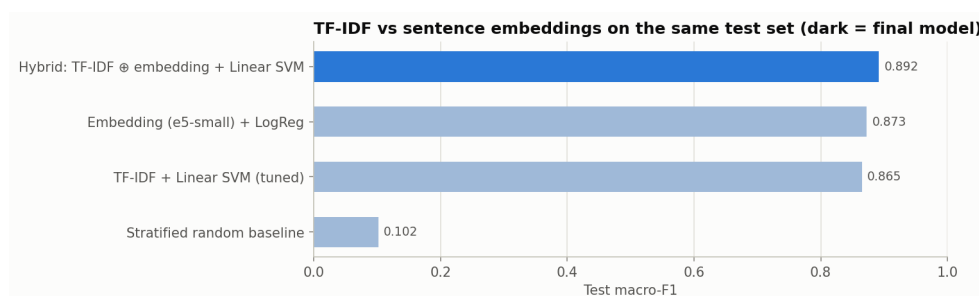
**Figure 3.** Two questions about the data itself, answered with the tuned TF-IDF + Linear SVM pipeline.

**Question.** Is the headline or the lead more informative — and is our crawled dataset big enough?

**Finding.** The **lead alone (0.844)** beats the **headline alone (0.790)** by +0.053: leads are longer and written to summarise, headlines are short and often figurative. Using both is best (0.857, +0.013 over the lead). The learning curve rises from 0.742 with 263 articles to 0.845 with 1,319, then flattens: doubling to 2,638 articles adds only +0.012. Crawling more would help a little; a better representation should help more.

### 6.2 TF-IDF versus multilingual sentence embeddings

We encode every article with `intfloat/multilingual-e5-small` [2], a 118-million-parameter multilingual sentence encoder with 384-dimensional output, loaded with the `sentence-transformers` library [3]. Following the assignment, the pretrained model is used as a **frozen feature extractor**: a Logistic Regression is trained on the embeddings. A **hybrid** model concatenates the TF-IDF features with the embeddings and trains a Linear SVM.



**Figure 4.** Test macro-F1 of the three approaches and the random baseline (dark = final model, chosen on validation).

**Question.** Does a pretrained neural encoder beat a bag of syllables on short Vietnamese news?

**Finding.** Yes, and combining them is best. On the test set, embeddings alone reach 0.873 (+0.007 over tuned TF-IDF at 0.865), and the **hybrid reaches 0.892** (+0.027). The ranking is the same on validation (0.857 → 0.882 → 0.888), so the choice generalises. TF-IDF keeps exact names (*iphone*, *asiad*) that embeddings blur; embeddings map different wordings of the same idea close together.

## 7 Results on the test set

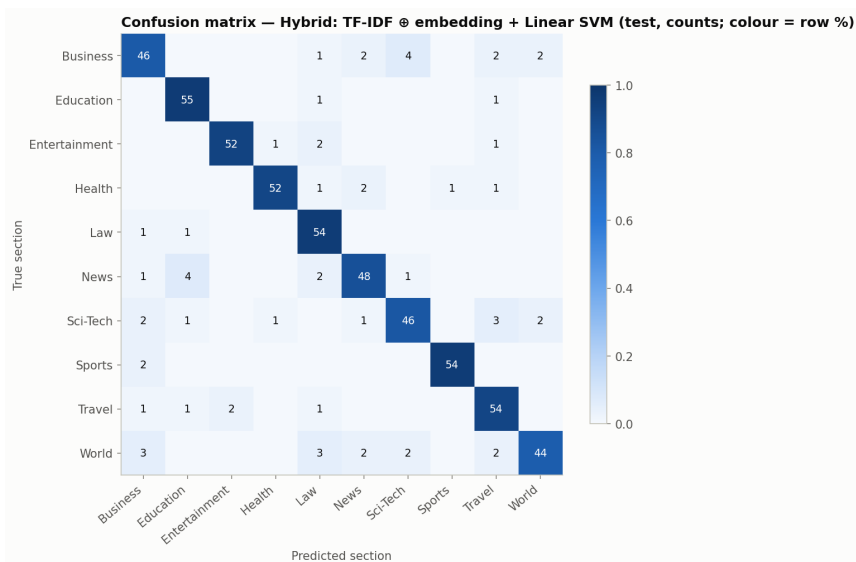
**Table 7.** Validation and test scores of the compared approaches. The final model was chosen on validation macro-F1.

| Approach                                       | Val. macro-F1 | Test macro-F1 | Test acc.    |
|------------------------------------------------|---------------|---------------|--------------|
| Random guess (stratified)                      | —             | 0.102         | —            |
| TF-IDF 1–3-gram + Linear SVM (tuned)           | 0.857         | 0.865         | 0.866        |
| Embedding (e5-small) + Logistic Regr.          | 0.882         | 0.873         | 0.873        |
| <b>Hybrid: TF-IDF ⊕ embedding + Linear SVM</b> | <b>0.888</b>  | <b>0.892</b>  | <b>0.892</b> |

The final hybrid model reaches **accuracy 0.892 and macro-F1 0.892** on the 566 test articles; 61 are misclassified. *Sports* is the easiest section (F1 0.973) and *Business* the hardest (0.814).

**Table 8.** Per-section test F1 of the final model.

| Section       | F1    | Section  | F1    |
|---------------|-------|----------|-------|
| Sports        | 0.973 | Travel   | 0.878 |
| Entertainment | 0.946 | News     | 0.865 |
| Health        | 0.937 | World    | 0.846 |
| Education     | 0.924 | Sci-Tech | 0.844 |
| Law           | 0.893 | Business | 0.814 |



**Figure 5.** Confusion matrix of the final model on the test set (counts; colour = share of the true section).

**Table 9.** Most frequent confusions on the test set.

| True section | Predicted as | Test errors |
|--------------|--------------|-------------|
| News         | Education    | 4           |
| Business     | Sci-Tech     | 4           |
| World        | Law          | 3           |
| Sci-Tech     | Travel       | 3           |
| World        | Business     | 3           |

**Question.** Did the centroid similarity of Figure 2 predict the errors?

**Finding.** Largely yes: **Business** → **Sci-Tech**, the most similar pair before training, is among the two most frequent confusions. Business is the hardest section because it overlaps with technology, world affairs and travel (airlines, tourism revenue).

**Table 10.** Examples of misclassified test articles.

| Headline (Vietnamese)                                                 | True          | Predicted |
|-----------------------------------------------------------------------|---------------|-----------|
| Thủ đô quốc gia Đông Nam Á nào đông dân nhất thế giới?                | Travel        | Education |
| Giải cứu 10 ngư dân Việt Nam làm việc tại Malaysia                    | World         | News      |
| Hàng không đón khách quốc tế nhiều hơn nội địa dịp Quốc khánh         | Business      | Travel    |
| Tại sao cửa sổ máy bay khó sản xuất?                                  | Sci-Tech      | Travel    |
| Nghệ thuật 'khiêu khích' ẩn sau quả chuối dán tường 6 triệu USD       | World         | Travel    |
| Nổi dậy dớt của đầu bếp thoát chết vụ 11/9 nhờ đổi ca với đồng nghiệp | World         | Law       |
| Justin Bieber mở show miễn phí vì người vô gia cư                     | Entertainment | Travel    |
| Mark Zuckerberg mua lâu đài                                           | Sci-Tech      | Travel    |

Many errors sit genuinely between sections — “*Tại sao cửa sổ máy bay khó sản xuất?*” (why aircraft windows are hard to make) is Sci-Tech but reads like Travel; a human editor could file several of them either way, so part of the remaining error is label ambiguity rather than model failure. Finally, all eight headlines written by the group, which are not in the dataset, are classified correctly:

**Table 11.** Predictions of the final model on new headlines.

| Headline written by the group (not in the dataset)      | Predicted section |
|---------------------------------------------------------|-------------------|
| Đội tuyển Việt Nam thắng Thái Lan 2-1 ở chung kết       | Sports            |
| Giá vàng tăng mạnh phiên đầu tuần                       | Business          |
| Bộ Giáo dục công bố phương án thi tốt nghiệp THPT 2027  | Education         |
| Phát hiện hành tinh mới có thể chứa nước                | Sci-Tech          |
| Bác sĩ khuyến cáo cách phòng sốt xuất huyết mùa mưa     | Health            |
| Khởi tố giám đốc công ty lừa đảo chiếm đoạt 50 tỷ đồng  | Law               |
| Tổng thống Mỹ gặp Chủ tịch Trung Quốc tại hội nghị APEC | World             |
| Vịnh Hạ Long lọt top điểm đến đẹp nhất châu Á           | Travel            |

## 8 Conclusions

1. **A self-crawled Vietnamese dataset is enough for a strong classifier:** 3,769 clean articles in 10 sections give 0.892 test macro-F1 from a headline and a two-sentence lead.
2. **Vietnamese needs care in the details:** NFC normalisation, keeping one-character syllables, and syllable bigrams to recover compound words. TF-IDF 1–2-grams beat raw unigram counts by +0.044 (Logistic Regression) and +0.044 (Linear SVM).
3. **Sentence embeddings beat TF-IDF, and the combination is best:** +0.007 and +0.027 test macro-F1 over tuned TF-IDF.
4. **The lead carries more signal than the headline** (0.844 vs 0.790), and the learning curve flattens after about 1,319 articles.
5. **Errors are predictable:** the most similar pair of section centroids is among the most frequent confusions.

**Limitations.** All articles come from one short window (the latest 12 listing pages per section on 27 September 2026), and many discriminative words are event names (Trump, ASIAD, iPhone, floods), so accuracy will likely drop on news from another period. There is one newspaper and one editorial style, and section labels are sometimes ambiguous. Syllable n-grams only approximate words; a Vietnamese word segmenter was not used, to keep the notebook dependency-free.

**Future work.** Re-crawl in a later month and measure the drop (temporal robustness); crawl other newspapers to test cross-source generalisation; compare word-segmented TF-IDF; lightly fine-tune a Vietnamese language model on top of the frozen-encoder baseline.

## References

- [1] VnExpress — Vietnamese online newspaper. <https://vnexpress.net> (section listings crawled 27 September 2026).
- [2] L. Wang, N. Yang, X. Huang, L. Yang, R. Majumder, F. Wei. *Multilingual E5 Text Embeddings: A Technical Report*. arXiv:2402.05672, 2024. Model: <https://huggingface.co/intfloat/multilingual-e5-small>
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- [5] Group G3. *Notebook G3\_Text\_VnExpressNews.ipynb*. [https://github.com/trungly-hcmut/C05177\\_DataVisualization\\_Assignment/blob/main/text/G3\\_Text\\_VnExpressNews.ipynb](https://github.com/trungly-hcmut/C05177_DataVisualization_Assignment/blob/main/text/G3_Text_VnExpressNews.ipynb)
- [6] Group G3. *Project page (rendered notebook, report and video)*. [https://trungly-hcmut.github.io/C05177\\_DataVisualization\\_Assignment/text/](https://trungly-hcmut.github.io/C05177_DataVisualization_Assignment/text/) — video: to be published